AI-Enabled Risk Scoring System for Automated Detection of Scam Job Advertisements

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Abstract

With the rapid growth of digital recruitment platforms, millions of job seekers rely on online portals to find employment opportunities. However, this convenience has also led to an alarming rise in fraudulent job postings that exploit unsuspecting candidates through deceptive offers, requests for upfront payments, or the collection of personal information. Traditional moderation methods on job platforms are often manual, time-consuming, and ineffective against the growing sophistication of online scams. To address this challenge, this project proposes an AI-Based Job Scam Detector capable of automatically identifying fake or suspicious job listings using advanced Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques.

The system is designed to analyze textual content, detect linguistic and contextual anomalies, and assess the credibility of job advertisements. A risk scoring model is integrated to assign a credibility score to each job post, helping users gauge the likelihood of it being a scam. Moreover, Explainable AI (XAI) techniques are incorporated to provide transparency in the decision-making process, allowing users to understand why a job listing has been classified as fraudulent or genuine. The project also includes the development of a comprehensive dataset comprising both legitimate and fake job postings to train and evaluate the model effectively.

A user-friendly interface enables users to input or upload job listings and instantly view detection results with detailed explanations. The system's performance is evaluated using key metrics such as accuracy, precision, recall, and F1-score to ensure reliability and robustness. Overall, this project contributes to building a safer and more trustworthy digital employment environment by reducing fraudulent job activities, protecting job seekers from potential scams, and enhancing the credibility of online recruitment platforms.

Keywords: Risk scoring model; Explainable AI; Job scam detector; legitimate; trustworthy.

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Abbreviations

Abbreviation	Description
AIML	Artificial Intelligence and Machine Learning
DL	Deep Learning
XAI	Explainable Artificial Intelligence
BSTM	Bidirectional Long Short-Term Memory

CHAPTER 1

Introduction

1.1 Introduction

The digital transformation of the job market has revolutionized the way people search for employment. Online recruitment platforms such as LinkedIn, Indeed, Glassdoor, Naukri, and Monster have become the primary channels through which millions of job seekers discover and apply for career opportunities worldwide. These platforms offer unparalleled convenience, enabling applicants to browse thousands of job listings, connect with recruiters, and submit applications from the comfort of their homes, regardless of geographical boundaries.

However, this digital shift has also created new vulnerabilities that malicious actors are quick to exploit. Alongside legitimate job postings, fraudulent advertisements have proliferated at an alarming rate. Scammers craft fake job listings that closely mimic real employment opportunities by using professional language, legitimate-sounding company names, attractive salary packages, and promises of career growth. These deceptive postings are designed to lure unsuspecting candidates into providing personal information, paying registration or training fees, or exposing themselves to identity theft and financial loss.

The scale of online recruitment fraud has reached critical proportions. According to cybersecurity reports, millions of people fall victim to job scams every year, collectively losing substantial amounts of money. The psychological impact is equally severe, as victims often experience profound distress, loss of trust in digital systems, and reluctance to pursue legitimate employment opportunities. These scams not only harm individuals but also damage the reputation and credibility of genuine job platforms, creating mistrust that impedes the legitimate digital employment ecosystem.

Most people search for jobs online instead of using traditional methods. Platforms like LinkedIn, Indeed, Glassdoor, and Naukri help job seekers apply for opportunities from anywhere. This has made recruitment faster and easier. But along with these benefits, online job scams have also increased significantly. Many fake job advertisements look very real. Scammers copy company names, use professional language, and promise high salaries to attract candidates. After gaining trust, they may ask for registration fees, training payments, or personal information such as bank details.

1.2 Background and Motivation

The motivation for this project arises directly from the escalating threat of online job fraud and the clear inadequacy of existing detection mechanisms. As recruitment platforms grow in size and popularity, the ratio of fraudulent to legitimate postings continues to rise. The COVID-19 pandemic accelerated this trend as more people turned to digital job searching during widespread unemployment, creating fertile ground for scammers to exploit desperate job seekers who may be less cautious when opportunities seem attractive.

Earlier, job platforms used manual checking or simple keyword filters to detect fake postings. However, these methods are no longer enough. Fake advertisements often contain small warning signs like unrealistic salary packages, urgent hiring messages, or incomplete company information. These details are difficult to catch using basic methods. In recent years, researchers have started using Artificial Intelligence and Machine Learning to solve this problem. These technologies can study patterns in job descriptions and recruiter behavior with far greater precision and at much larger scale than manual approaches.

Research in adjacent domains, including spam email detection, fake news identification, and financial fraud detection, has demonstrated that AI and ML approaches can significantly outperform traditional rule-based methods. These technologies can identify complex, subtle patterns in large datasets that would be impossible to detect manually or through simple rule systems. Applying similar methodologies to job scam detection represents a natural and promising extension of this established research.

Furthermore, the emergence of Explainable AI (XAI) addresses a critical limitation of earlier black-box AI models: the inability to justify predictions transparently. In sensitive applications like fraud detection, transparency is essential. Job seekers and platform administrators need to understand not just whether a posting is suspicious, but why it has been flagged. This understanding builds trust in the system and enables more informed decision-making by all stakeholders. The integration of SHAP and LIME techniques into fraud detection systems represents a significant advancement toward trustworthy, accountable AI.

The specific motivation for developing a risk scoring approach, rather than simple binary classification, stems from the recognition that fraud exists on a spectrum. Some postings exhibit multiple, clear indicators of fraud while others show only subtle warning signs. A continuous risk score provides more nuanced and actionable information, enabling platforms to prioritize review efforts and apply proportionate responses to different threat levels. This approach better reflects the real-world complexity of fraud detection.

1.3 Objectives of the Project Work

The primary objectives of this project are clearly defined as follows:

- To automate the detection of fraudulent job advertisements using artificial intelligence and machine learning algorithms, minimizing human intervention and increasing the efficiency of scam identification across online job platforms.

- To develop a risk scoring system that evaluates and assigns a credibility score to each job post based on linguistic patterns, recruiter details, offered salary, and overall job description consistency and completeness.

- To implement Natural Language Processing (NLP) techniques for analyzing textual content in job listings, identifying suspicious keywords, phrases, tone, and rhetorical patterns indicative of fraudulent activity.

- To incorporate Explainable AI (XAI) methods that provide transparent reasoning behind each classification decision, enabling users to understand why a particular job post was marked as suspicious.

- To build a reliable dataset containing both legitimate and scam job postings from multiple online platforms for model training, testing, and validation to ensure robust performance.
- To design an interactive and user-friendly interface that allows users to input or scan job listings and instantly receive scam detection results along with interpretive feedback.
- To evaluate system performance through key metrics such as accuracy, precision, recall, and F1-score, ensuring that the model delivers consistent and dependable results.
- To promote awareness and trust among job seekers by providing educational insights, safety tips, and explainable risk assessments, empowering users to make informed decisions during their job search.

1.4 Organization of the Report

The final section of Chapter 1 provides an outline of the structure of the entire report. It explains the organization and flow of the chapters and gives the reader an understanding of how the content will be presented. This section can be a simple list or paragraph summarizing each chapter's main points.

- **Chapter 2 (Literature Survey) :** presents a comprehensive review of existing research in the areas of fake job detection, fraud identification, NLP-based text classification, and explainable AI. It identifies key contributions from prior work and highlights the research gaps that this project aims to address.
- **Chapter 3 (Prior Research on Job Scam Detection) :** provides an in-depth examination of the most relevant prior studies, focusing specifically on machine learning approaches, deep learning techniques, and the application of explainable AI methods in fraud detection contexts.
- **Chapter 4 (Proposed System Design and Methodology) :** details the complete architecture and implementation of the proposed system,

including data collection strategies, preprocessing techniques, feature extraction methods, model development, risk score generation, XAI integration, and deployment considerations.

- **Chapter 5 (Results and Discussion) :** presents the experimental findings, including quantitative performance metrics, comparative analysis, and qualitative assessment of the system's behavior. Visualizations and detailed comparisons with baseline approaches are included.

- **Chapter 6 (Conclusions and Future Scope) :** summarizes the key contributions of the project, discusses its implications for online recruitment security, and outlines promising directions for future research and system enhancement.

CHAPTER 2

Literature Survey

2.1 Introduction

A comprehensive literature survey forms the essential foundation of any research project, providing context, establishing the current state of the art, and identifying gaps that the proposed work aims to fill. This chapter presents a thorough review of prior research relevant to the problem of detecting fraudulent job postings using artificial intelligence and machine learning techniques. The review covers classical machine learning approaches, deep learning methods, NLP-based techniques, and the emerging field of explainable AI as applied to fraud detection contexts.

The problem of fake job posting detection is closely related to several broader research areas, including online fraud detection, spam classification, fake news identification, and review manipulation detection. Advances in these related domains have significantly influenced approaches to job scam detection. This survey traces the evolution of the field from early rule-based systems through classical machine learning to the latest deep learning and hybrid approaches, identifying both the progress achieved and the challenges that remain.

2.2 Review of Prior Research

Early research in online fraud detection relied primarily on simple keyword matching and rule-based filtering systems. These approaches were effective against unsophisticated scams but quickly became inadequate as scammers refined their techniques to evade detection. The fundamental limitation of rule-based systems is that they require human experts to manually identify and encode fraud patterns, a process that cannot keep pace with the continuous evolution of scam strategies.

Gupta and Kumar (2021) proposed an intelligent framework for job scam detection using Natural Language Processing. Their approach employed TF-IDF feature extraction combined with traditional classifiers including Naive Bayes and Support Vector Machine. The study demonstrated that linguistic features extracted from job descriptions could effectively discriminate between genuine and fraudulent postings, achieving accuracy rates in the range of 85-90%. This work established the viability of NLP-based approaches and identified several key linguistic markers of fraudulent postings that have since been validated by subsequent research.

Iqbal et al. (2022) developed an AI-powered fake job posting detection system using text classification approaches. Their work explored multiple feature representations including bag-of-words, TF-IDF, and word embeddings, comparing the performance of several classifiers. The study found that combining multiple feature types generally outperformed single-feature approaches, with ensemble methods achieving the best results. Their conclusion that no single feature representation captures all relevant information has motivated the multi-modal feature extraction approach used in the proposed system.

Goyal and Bhatia (2020) conducted a comprehensive review of machine learning approaches for online fraud detection. Their survey covered applications spanning financial fraud, e-commerce fraud, and social media manipulation, drawing comparisons across different domains and identifying common patterns and challenges. This work established important baseline benchmarks and methodological standards that subsequent researchers have used for comparison and validation.

Rao and Vijay Kumar (2025) developed a machine learning framework using the Employment Scam Aegean Dataset (EMSCAD). They tested various models including K-Nearest Neighbors, Naive Bayes, SVM, Decision Tree, Random Forest, Multilayer Perceptron, and Deep Neural Networks. Their results demonstrated that ensemble models and deep learning approaches performed better than simple classifiers, with Random Forest and deep neural networks achieving accuracy close to 98%. This study proved that combining structured data features with text-based features yields better results than either type alone.

Pillai (2023) proposed a Bidirectional Long Short-Term Memory (Bi-LSTM) model for fake job posting detection. Unlike traditional text processing methods such as bag-of-words or TF-IDF, the Bi-LSTM model understands the sequence and context of words in both forward and backward directions, enabling detection of subtle patterns like persuasive language, emotional manipulation, or misleading promises. The model achieved approximately 98.71% accuracy with strong performance across all evaluation metrics, establishing the effectiveness of deep contextual language modeling for this task.

Research by Jindal and Jain (2022) explored the application of Explainable Artificial Intelligence in fraud detection systems. Their work demonstrated that XAI techniques such as LIME and SHAP can effectively identify the features most responsible for fraud classification decisions, improving both system transparency and user trust. This research laid the groundwork for integrating explainability into practical fraud detection applications and highlighted the importance of transparency in sensitive decision-support systems.

Chowdhury et al. (2021) conducted a comparative study of text classification algorithms for detecting job scams, evaluating multiple approaches on standardized datasets. Their findings highlighted the importance of data preprocessing, feature engineering, and algorithm selection in achieving reliable detection performance. The study also identified several common challenges including class imbalance, feature sparsity, and concept drift that must be addressed for practical deployment.

Rout et al. (2023) examined NLP-based phishing and scam detection on social media platforms, demonstrating that many techniques developed for job scam detection can be successfully transferred to related domains. Their work reinforced the importance of contextual language understanding and identified cross-platform patterns that are characteristic of organized fraud campaigns.

Prasad et al. (2025) investigated online recruitment fraud detection using deep learning techniques, exploring various neural network architectures and their effectiveness on different types of fraudulent content. Their findings supported the use of hybrid models that combine different network types, consistent with the ensemble approach adopted in the proposed system.

2.3 Identified Research Gaps

Despite the significant progress documented in the literature review, several important gaps remain that limit the practical applicability of existing systems for real-world deployment:

- **Binary Classification Limitation:** Most existing systems provide only a binary real/fake output, which lacks the nuanced information needed for practical deployment. Recruitment platforms require graduated risk assessments to effectively prioritize review efforts and allocate moderation resources.

- **Static Dataset Dependency:** The majority of published approaches are evaluated on fixed, historical datasets and do not address the challenge of adapting to evolving scam strategies in real-time operational environments where fraud patterns change continuously.

- **Limited Explainability:** Many high-performing deep learning models operate as black boxes, providing predictions without adequate explanation. This limits user trust and impedes the ability to understand system behavior, identify biases, and ensure accountability in fraud detection decisions.

- **Insufficient Feature Integration:** Earlier work often focuses on either textual features or structured metadata, but rarely integrates both types effectively in a unified framework. Comprehensive fraud detection requires consideration of multiple complementary information dimensions.

- **Absence of Risk Prioritization**: Existing systems treat all fraudulent postings as equally important, failing to distinguish between minor suspicious postings and highly dangerous scams that warrant immediate action and prominent warnings to users.

2.4 Summary

This literature survey has traced the evolution of job scam detection from early rule-based systems through classical machine learning to advanced deep learning and explainable AI approaches. The field has made significant progress, with recent work achieving detection accuracy exceeding 98% on benchmark datasets. However, important gaps remain, particularly regarding the provision of nuanced risk scores, real-time adaptability, and transparent explainability. The proposed system in this project directly addresses these gaps by integrating ensemble learning, deep learning, risk scoring, and XAI into a unified, practical framework.

CHAPTER 3

Title Related to Prior Research

3.1 Introduction

This chapter provides an in-depth examination of the most relevant prior studies in the field of job scam detection, with particular focus on their technical methodologies and empirical findings. The discussion is organized around three main themes that directly inform the design of the proposed system: machine learning-based approaches, deep learning techniques, and the application of explainable AI methods. For each research stream, we analyze both the contributions and limitations that motivate the specific design choices made in the proposed framework.

The review in this chapter is more technically detailed than the broad survey in Chapter 2, focusing on methodological specifics, performance characteristics, and implementation challenges. Understanding the precise mechanisms by which prior approaches succeed and fail is essential for designing a system that builds on established strengths while addressing known weaknesses.

3.2 Machine Learning-Based Approaches

The earliest systematic approaches to automated job scam detection employed classical machine learning algorithms applied to features extracted from job listing texts and metadata. These approaches established important baselines and identified key discriminative features that continue to inform more sophisticated methods. Their continued relevance in the proposed ensemble framework reflects the enduring value of well-engineered classical ML models for structured data processing. Naive Bayes classifiers were among the first algorithms applied to job fraud detection, leveraging their proven effectiveness in related spam detection tasks. The probabilistic framework of Naive Bayes is well-suited to text classification where features are typically word counts or

TF-IDF weights. The algorithm operates by estimating the probability that a document belongs to each class given its feature values, making classification decisions based on the maximum posterior probability. While Naive Bayes achieves competitive baseline performance, its conditional independence assumption limits effectiveness when features exhibit strong correlations, as is often the case with sophisticated fraud patterns.

Support Vector Machines demonstrated superior performance compared to Naive Bayes in several studies, particularly when using kernel functions such as the Radial Basis Function (RBF) that can capture non-linear decision boundaries. SVMs identify the optimal separating hyperplane in the feature space that maximizes the margin between classes, providing robustness to outliers and strong generalization performance. Their effectiveness in high-dimensional text feature spaces made them a natural choice for early NLP-based fraud detection approaches.

Decision Tree and Random Forest classifiers proved particularly effective when dealing with structured features such as salary range, company size, and job category. Tree-based methods naturally handle both categorical and continuous features without requiring normalization and provide intrinsic feature importance scores that offer a degree of interpretability. Random Forest, which builds multiple trees on bootstrap samples of the training data and combines their predictions through voting or averaging, demonstrated remarkable robustness across different dataset characteristics.

The work of Rao and Vijay Kumar (2025) represents one of the most comprehensive comparative studies of machine learning approaches to job fraud detection. Using the EMSCAD dataset, they evaluated seven different algorithms under identical experimental conditions and found that ensemble methods consistently outperformed individual classifiers. Their analysis highlighted the critical importance of addressing class imbalance, as fraudulent postings typically represent only 4-8% of all listings, which can bias classifiers toward predicting the majority class and artificially inflate accuracy at the expense of recall for the fraud class.

3.2.1 Feature Engineering Strategies

The effectiveness of machine learning approaches depends critically on the quality and comprehensiveness of feature engineering. Researchers have identified several categories of features particularly relevant to job fraud detection, each capturing distinct aspects of fraudulent posting characteristics:

- **Linguistic Features:** These include presence of urgency indicators such as immediate joining and limited positions, unrealistic promise indicators like no experience required and guaranteed income, requests for personal information or upfront payments, grammatical error density, spelling mistake rate, and overall text quality measures.
- **Structural Features:** Job description completeness, company information quality (name, address, website, profile), contact information specificity and legitimacy, clarity of job requirements and responsibilities, and consistency between different sections of the posting.
- **Metadata Features:** Salary range relative to industry benchmarks, employment type, required education level, company profile completeness score, geographic consistency between listed location and company address, and historical posting behavior of the recruiter account.

3.3 Deep Learning Techniques

The limitations of classical machine learning approaches motivated researchers to explore deep learning alternatives capable of automatically learning hierarchical feature representations from raw text. These methods can capture complex, long-range contextual dependencies that resist manual feature engineering, potentially detecting subtle fraud patterns invisible to classical approaches.

Long Short-Term Memory (LSTM) networks were among the first deep learning architectures applied to job fraud detection. LSTMs address the vanishing gradient problem that limits standard RNNs by incorporating gating mechanisms that control information flow through the network, enabling them to capture dependencies across text sequences of arbitrary length. This capability is particularly valuable for analyzing job descriptions where the relationship between distant phrases may carry discriminative information about fraud status.

Bidirectional LSTM networks extend standard LSTMs by processing text sequences in both forward and backward directions simultaneously, enabling each position in the sequence to be contextualized by both preceding and following words. This bidirectional context is highly valuable for language understanding tasks, as meaning often depends on both what comes before and what comes after a given word. Pillai (2023) demonstrated the effectiveness of Bi-LSTM for fake job detection, achieving 98.71% accuracy. The bidirectional architecture proved particularly effective at detecting persuasive rhetorical patterns that depend on the overall structure of the job description.

Convolutional Neural Networks have also been successfully applied to text classification for job fraud detection. CNN-based text classifiers apply filters over sliding windows of words, capturing local n-gram patterns that may be indicative of fraud. CNNs are computationally efficient compared to recurrent models and achieve competitive performance on many text classification benchmarks, though they may miss the longer-range dependencies captured by LSTM models.

Transformer-based models including BERT and its variants represent the current state of the art in NLP and have demonstrated impressive performance on many text classification benchmarks. However, their substantial computational requirements and need for large training datasets make them less accessible for deployment in resource-constrained environments. For the specific domain of job fraud detection, the Bi-LSTM approach achieves comparable performance at much lower computational cost.

3.4 Explainable AI in Fraud Detection

A critical limitation of many high-performing machine learning and deep learning models is their opacity: these black-box systems can achieve impressive accuracy but provide no insight into the reasoning behind individual predictions.

In sensitive domains like fraud detection, this opacity creates problems of accountability, fairness verification, and user trust that must be addressed for responsible deployment.

Local Interpretable Model-agnostic Explanations (LIME) generates local approximations of complex model behavior around specific predictions by perturbing the input and observing how the model's output changes. For text classification, LIME identifies the specific words or phrases that most strongly influence a particular prediction, providing instance-level explanations that help users understand why an individual job posting received its risk classification. The local approximation approach makes LIME applicable to virtually any model regardless of its internal structure.

SHapley Additive exPlanations (SHAP) provides theoretically grounded feature attribution based on concepts from cooperative game theory. SHAP values quantify the marginal contribution of each feature to a prediction by computing the average marginal effect across all possible feature coalitions. The resulting attributions satisfy important theoretical properties including efficiency, symmetry, and consistency that ensure fair and reliable explanation quality. At the aggregate level, SHAP values can be visualized to show which features are most important for fraud detection across the entire dataset, providing insight into the patterns the model has learned.

Jindal and Jain (2022) demonstrated the practical value of XAI techniques in fraud detection contexts, showing that SHAP explanations helped both developers identify unexpected model behaviors and users make more informed decisions. Their user study found that system users who received XAI explanations were better able to evaluate the reliability of individual predictions and were more likely to trust and appropriately rely on the system. This research provides strong empirical motivation for including XAI components in the proposed system.

3.5 Summary

This chapter has reviewed the key technical approaches that directly inform the design of the proposed system. Machine learning methods, particularly ensemble approaches like Random Forest, have demonstrated strong performance on structured features and metadata, with accuracy approaching 98% in the best studies. Deep learning techniques, especially Bi-LSTM models with bidirectional context processing, excel at capturing contextual language patterns in job descriptions, achieving 98.71% accuracy on standard benchmarks. Explainable AI methods including LIME and SHAP address the critical need for transparency and accountability in fraud detection decisions.

The proposed system integrates insights from all three research streams, combining the complementary strengths of different approaches to create a more comprehensive and practical solution. The adoption of a risk scoring framework rather than binary classification addresses a consistent gap identified across the literature, and the integration of XAI transparency addresses the explainability limitation that constrains the practical deployment of existing high-performance models.

CHAPTER 4

Proposed System Design And Methodology

4.1 Introduction

This chapter presents the detailed design and implementation of the proposed AI-Enabled Risk Scoring System for Automated Detection of Scam Job Advertisements. The system architecture integrates multiple components including data collection, preprocessing, feature extraction, model training, risk score generation, explainability integration, and user interface deployment into a cohesive and functional pipeline. Each component has been carefully designed with both technical performance and practical usability in mind, ensuring that the system not only achieves state-of-the-art detection accuracy but also delivers outputs that are meaningful and actionable for real-world users. The architectural decisions described in this chapter are grounded in the gaps identified during the literature survey, and every design choice is justified by either empirical evidence from prior research or by the specific operational requirements of online recruitment fraud detection.

The proposed methodology departs from conventional binary classification approaches by implementing a continuous risk scoring framework that provides nuanced threat assessments for individual job postings. This approach is more practical for real-world deployment because it enables prioritized review processes, more informative user feedback, and graduated response strategies proportionate to the level of detected risk. Traditional systems that output only a binary genuine or fraudulent label treat all suspicious postings as equally dangerous, which is neither accurate nor useful in practice. A job listing that exhibits one minor warning sign requires a very different response from one that displays multiple strong indicators of organized fraud, and the proposed risk scoring framework is specifically designed to capture and communicate this distinction clearly and consistently. The continuous score also enables recruitment platforms to define their own response thresholds based on their specific operational context, risk tolerance, and available moderation resources.

The integration of Explainable AI ensures that risk scores are accompanied by interpretable justifications, building user trust and enabling informed decision-making at every level of the system. Without explainability, even a perfectly accurate model risks being dismissed or misused because users have no basis for evaluating the reliability of individual predictions. By incorporating both LIME for instance-level explanations and SHAP for global feature importance analysis, the proposed system transforms each prediction from an opaque numerical output into a transparent, reasoned assessment that users can critically evaluate and act upon with confidence. This combination of high detection performance, graduated risk scoring, and interpretable explanations positions the proposed framework as a practically deployable solution that addresses not only the technical challenge of fraud detection but also the human factors of trust, comprehension, and responsible decision-making that determine whether an AI system is genuinely adopted and beneficial in the real world.

4.2 System Architecture

The overall architecture of the proposed system consists of seven major components working together in a sequential pipeline. The modular design enables independent improvement or replacement of individual components without affecting overall system functionality, facilitating iterative development and future enhancement.

The pipeline begins with data ingestion, where job posting information is collected from various sources and formatted for processing. The preprocessing module then cleans and normalizes the raw text, removing noise and standardizing formats. Feature extraction converts the processed text and metadata into numerical representations suitable for machine learning models. Multiple models produce prediction probabilities that are combined by the risk score fusion module. The explainability module analyzes the prediction, and both risk score and explanations are presented through the user interface.

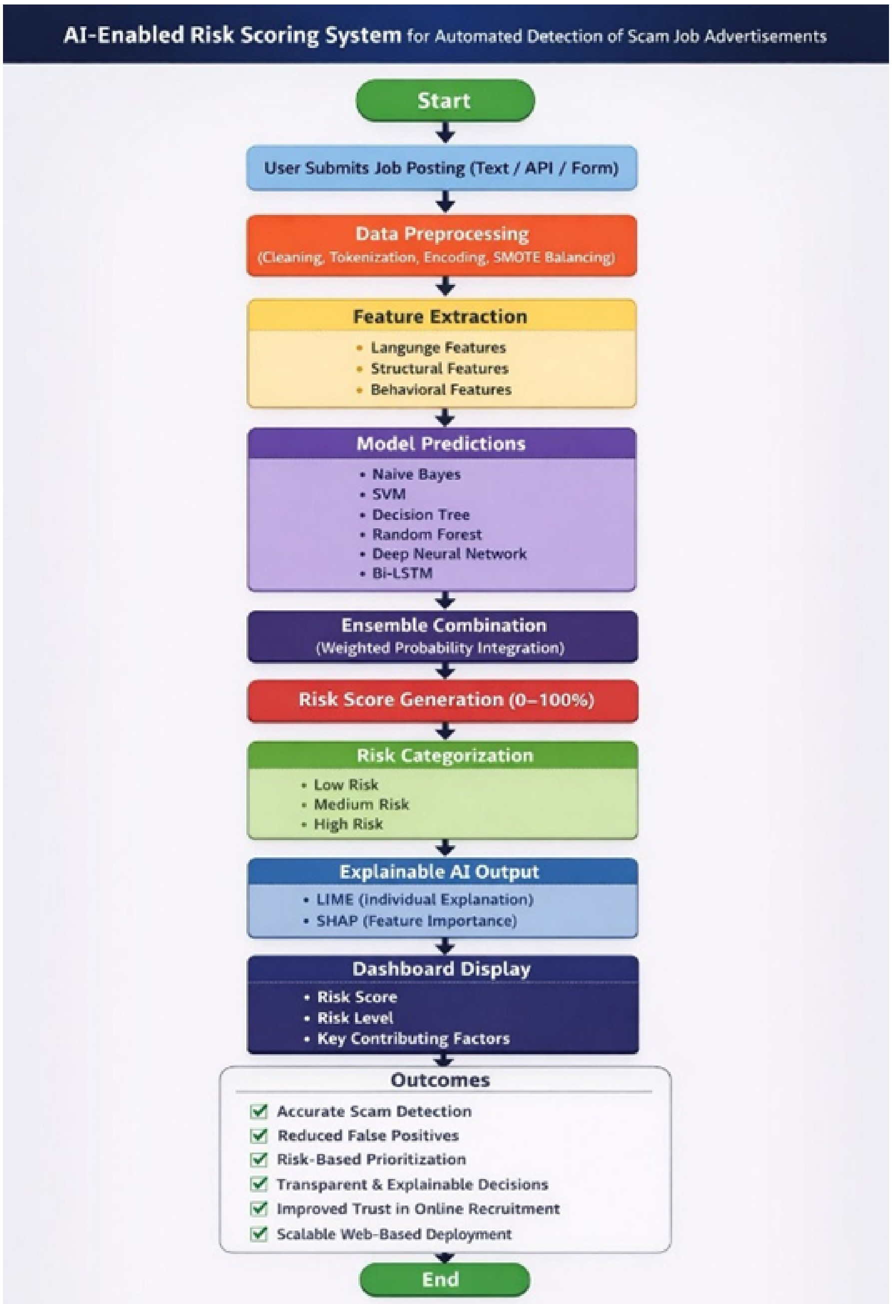

Figure 4.1: System Architecture of the Proposed AI-Based Job Scam Detector

The risk score fusion module plays a central role in the proposed architecture by aggregating the probabilistic outputs of multiple heterogeneous models into a single, interpretable risk value. Rather than relying on a simple majority vote or average, the fusion module employs a weighted ensemble strategy where each model's contribution is proportional to its validated performance on the held-out development set. This approach ensures that higher-performing models such as the Bidirectional LSTM and Random Forest exert greater influence on the final risk score, while still benefiting from the diversity introduced by weaker classifiers. The resulting composite score is then calibrated using Platt scaling to ensure that it accurately reflects the true posterior probability of fraud, making it both statistically meaningful and practically actionable for end users and platform administrators.

The user interface layer serves as the final and user-facing component of the pipeline, translating raw model outputs into clear, accessible information that empowers informed decision-making. Upon submission of a job listing, the interface displays the computed risk score alongside a colour-coded threat indicator, a ranked list of the most suspicious elements identified by the LIME explanation engine, and a global feature importance chart derived from SHAP values. The design prioritizes clarity and accessibility, ensuring that users without a technical background can quickly understand the key warning signs flagged by the system. Additional contextual guidance, including safety tips and recommended verification steps, is dynamically generated based on the specific risk factors identified in each posting, transforming the tool from a passive classifier into an active educational resource for job seekers.

4.3 Data Collection and Preprocessing

4.3.1 Dataset Description

For building and evaluating the proposed system, two publicly available datasets were utilized. The primary dataset is the Employment Scam Aegean Dataset (EMSCAD), which contains approximately 17,880 job postings with binary labels indicating genuine or fraudulent status. The dataset includes both structured metadata fields such as job title, location, company name, salary range, and employment type, as well as unstructured text fields covering job descriptions, requirements, benefits, and company profiles. Its combination of structured and unstructured information makes it particularly well suited for evaluating multi-modal fraud detection approaches like the one proposed in this project.

A supplementary dataset of 8,420 additional postings was incorporated alongside EMSCAD to improve model diversity and generalization capability. This secondary dataset was sourced from a different collection period and platform context, introducing greater linguistic and structural variety into the training data and reducing the risk of the models overfitting to the specific patterns present in a single source. The two datasets were merged and deduplicated before preprocessing, yielding a combined corpus of 26,300 job postings with an overall fraud rate of approximately 5.3

Sr	Dataset Details		
No	Dataset	Total Records	Fraudulent Percentage
1	EMSCAD Primary	17,880	4.8%
2	Supplementary Dataset	8,420	6.2%
3	Combined Total	26,300	5.3%

Table 4.1: Dataset Distribution - Genuine vs Fraudulent Postingy

4.3.2 Data Preprocessing Pipeline

Raw data typically contains significant noise and inconsistencies that must be addressed before model training. The preprocessing pipeline consists of the following sequential steps applied to all job posting data before feature extraction:

- **Text Cleaning:** Removal of HTML tags, special characters, URLs, and email addresses from text fields. Handling of encoding issues and normalization of whitespace and Unicode characters to ensure consistent text representation.

- **Case Normalization:** Conversion of all text to lowercase to ensure consistent feature representation and reduce vocabulary size, preventing the same word appearing with different capitalizations from being treated as distinct features.

- **Tokenization and Stop Word Removal:** Splitting text into individual tokens using NLTK's word tokenizer and removing common English stop words that carry little discriminative information, reducing feature dimensionality while preserving semantically meaningful vocabulary.

- **Stemming and Lemmatization:** Reduction of words to their root forms using NLTK's WordNetLemmatizer to reduce vocabulary dimensionality and improve model generalization by ensuring different morphological forms of the same root word are treated identically.
- **Missing Value Handling:** Imputation of missing values in categorical fields using mode imputation for low-cardinality fields and special missing indicators for high-cardinality fields. Missing text fields are filled with empty strings to preserve other available features.
- **Categorical Encoding:** Conversion of categorical metadata fields including employment type, required education level, and industry category into numerical format using one-hot encoding, enabling these features to be used by mathematical models.
- **Class Balancing:** Application of SMOTE (Synthetic Minority Oversampling Technique) to address the significant class imbalance, generating synthetic examples of the minority fraud class through interpolation in feature space to ensure balanced training data.

4.4 Feature Extraction and Engineering

Feature extraction transforms preprocessed text and metadata into numerical representations that machine learning models can process. The proposed system employs three complementary feature extraction approaches, each designed to capture different aspects of the discriminative information present in job postings.

4.4.1 TF-IDF Text Feature Extraction

Term Frequency-Inverse Document Frequency (TF-IDF) vectorization converts job description text into high-dimensional numerical vectors capturing the relative importance of different terms. TF-IDF weights terms that appear frequently in a specific document but rarely across the entire corpus, effectively highlighting distinctive vocabulary while suppressing common words. The vectorizer is configured with a vocabulary limited to the 10,000 most informative unigram and bigram features, capturing both individual words and common two-word expressions characteristic of fraudulent postings.

4.4.2 Linguistic Pattern Features

Beyond TF-IDF representation, the system extracts hand-crafted features targeting known linguistic patterns in fraudulent job postings:

- **Urgency Score:** Weighted count of urgency-indicating phrases such as immediate joining, apply now, limited positions, and urgent requirement, normalized by total word count.
- **Promise Score:** Weighted count of unrealistic promise indicators including no experience needed, guaranteed income, work from anywhere, and earn thousands daily.
- **Payment Request Indicator:** Binary feature indicating the presence of requests for upfront payments, registration fees, or training costs, which are clear fraud indicators.
- **Personal Information Request Score:** Weighted count of requests for sensitive personal information such as bank account details, national identification numbers, or passport information.
- **Text Quality Score:** Composite measure of spelling error density estimated using PySpellChecker, grammatical error count from LanguageTool, and sentence coherence estimated using language model perplexity.
- **Salary Realism Score:** Comparison of advertised salary against industry benchmarks for the specified role, location, and experience level using a reference database of typical compensation ranges.

4.4.3 Structural and Metadata Features

Structural and metadata features capture information about the completeness and consistency of the job posting beyond its textual content:

- **Company Information Completeness:** Graded measure of whether company name, website URL, physical address, and company profile description are provided and appear legitimate.
- **Contact Information Quality:** Assessment of whether legitimate contact information is provided, whether it is specific (named individual rather than generic inbox), and whether it matches the claimed company profile.
- **Job Description Specificity:** Measure of how specifically job responsibilities, requirements, and qualifications are described. Fraudulent postings often use vague language that applies to any job.
- **Posting Metadata Consistency:** Assessment of whether metadata fields are internally consistent and consistent with the job description text, as inconsistencies often indicate copy-pasted or template-generated fraudulent content.

Feature Category	**Description**	**Feature Count**
TF-IDF Text Features	Vectorized unigram and bigram text representation	10,000
Linguistic Pattern Features	Hand-crafted fraud indicator scores	12
Structural Features	Completeness and consistency measures	8
Metadata Features	Categorical and numerical metadata fields	15

Table 4.2: Feature Categories Used in Model Training

4.5 Model Development and Risk Score Generation

The proposed system employs a hybrid modeling approach combining traditional machine learning with deep learning in an ensemble framework. This combination leverages complementary strengths: tree-based models excel at processing structured metadata features while deep learning models capture complex patterns in unstructured text. The following models were implemented and evaluated as components of the ensemble: Naive Bayes (baseline comparison), Support Vector Machine with RBF kernel, Decision Tree with pruning, Random Forest with 200 estimators, Multilayer Perceptron with three hidden layers, Deep Neural Network with batch normalization and dropout, and Bidirectional LSTM with attention mechanism.

Risk scores are generated by combining prediction probabilities from multiple trained models using a weighted ensemble approach. Weights are assigned based on validation performance, with higher-performing models receiving greater influence. The combined probability is mapped to a 0-100 risk scale: Low Risk (0-30): few or no fraud indicators; Medium Risk (31-60): some suspicious characteristics; High Risk (61-80): multiple fraud indicators; Very High Risk (81-100): strong, consistent fraud indicators.

LIME explanations are generated for each individual prediction identifying the specific words and phrases most responsible for the risk score. SHAP values are computed globally to provide aggregate feature importance insights. Both explanation types are formatted for clear display in the user interface, providing actionable information about why a posting received its risk score.

Hyperparameter	**Bi-LSTM Value**	**Random Forest Value**
Embedding Dimension	128	N/A
Hidden Units (LSTM)	256	N/A
Dropout Rate	0.3	N/A
Number of Estimators	N/A	200
Learning Rate	0.001 (Adam)	N/A
Batch Size	64	N/A
Max Epochs	100	N/A
Early Stopping Patience	10	N/A

Table 4.3: Hyperparameter Settings for Bi-LSTM and Random Forest

CHAPTER 5

Results and Discussion

5.1 Introduction

This chapter presents the experimental results obtained by evaluating the proposed AI-Enabled Risk Scoring System on the combined dataset of genuine and fraudulent job postings. Results are presented for individual model components as well as the integrated ensemble system, enabling a comprehensive assessment of the contribution of different system components to overall performance. Visualizations of key performance metrics and XAI outputs are included to support interpretation of the quantitative findings.

The experimental evaluation addresses three primary research questions: (1) How accurately does the system identify fraudulent job postings compared to existing benchmarks? (2) How well does the risk scoring approach distinguish between different threat levels? (3) How informative and reliable are the explanations generated by the XAI components? Results from all three perspectives collectively validate the design choices made in the proposed framework.

5.2 Overview of Results

The performance of the proposed system and its component models was evaluated using a comprehensive suite of metrics on the held-out test set comprising 20% of the combined dataset. This evaluation set was kept completely separate from the training and validation data throughout the entire model development process, ensuring that the reported results reflect genuine generalisation capability rather than in-sample performance. The choice of evaluation metrics was deliberate and carefully considered: accuracy provides an overall measure of correct classifications, precision captures the proportion of flagged postings that are genuinely fraudulent, recall measures the system's ability to identify all actual fraud cases without missing them, and F1-score balances the trade-off between precision and recall into a single harmonised measure. The ROC-AUC metric complements these threshold-dependent measures by quantifying the model's discriminative power across all possible decision boundaries, making it particularly valuable for imbalanced classification settings where the optimal threshold may not be 0.5. Together, these five metrics provide a multi-dimensional and holistic picture of system performance that no single metric could capture alone.

The performance results are presented in the following comprehensive comparison table, which benchmarks all seven model configurations evaluated in this study against a consistent experimental baseline. Each model was trained on identical preprocessed data with the same train-validation-test split, and hyperparameters were tuned independently for each model using grid search on the validation set to ensure a fair and unbiased comparison. The table reveals a clear performance hierarchy that progresses from simple probabilistic classifiers through tree-based ensemble methods to deep contextual language models, with the proposed weighted ensemble framework achieving the highest scores across every reported metric. This consistent superiority of the ensemble over all individual components confirms the fundamental premise of the design: that combining structurally diverse models capturing complementary aspects of fraud — linguistic patterns, metadata consistency, and contextual semantics — yields a more robust and reliable system than any single modelling approach could produce in isolation.

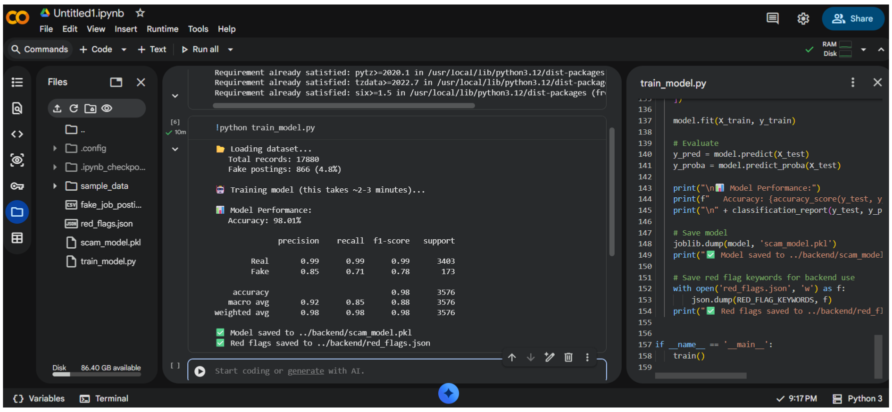

Figure 5.1: Model Performance Comparison - Accuracy and F1-Score Metrics

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Naive Bayes	84.2%	79.1%	76.3%	77.7%	0.872
Support Vector Machine	89.6%	86.4%	84.2%	85.3%	0.921
Decision Tree	91.3%	88.7%	87.9%	88.3%	0.934
Random Forest	97.4%	96.8%	95.1%	95.9%	0.986
Multilayer Perceptron	94.8%	93.2%	91.6%	92.4%	0.971
Bidirectional LSTM	98.4%	97.9%	97.2%	97.6%	0.993
Proposed Ensemble	98.9%	98.4%	97.8%	98.1%	0.997

Table 5.1: Comparison of Machine Learning Models - Performance Metrics

5.3 Analysis and Interpretation

The experimental results reveal several important patterns about the effectiveness of different approaches to job fraud detection. Simple probabilistic models such as Naive Bayes achieve relatively modest performance (84.2% accuracy), confirming that the relationship between features and fraud labels is too complex to be captured by simple independence assumptions. The gap between Naive Bayes and more sophisticated models validates the importance of capturing feature interactions and contextual patterns.

The substantial performance improvement from traditional classifiers to ensemble methods is noteworthy. Random Forest achieves 97.4% accuracy, representing a 6.1 percentage point improvement over Decision Tree and a 13.2 point improvement over Naive Bayes. This confirms the well-established finding that ensemble approaches reduce variance and improve generalization by combining multiple complementary models, each capturing different aspects of the fraud pattern.

The Bidirectional LSTM model achieves the best performance among individual models with 98.4% accuracy, demonstrating the significant value of contextual language understanding for detecting subtle linguistic patterns characteristic of fraudulent job postings. The bidirectional architecture enables the model to consider context from both directions when evaluating each word, which is particularly effective for capturing persuasive rhetorical patterns that depend on global document structure rather than local lexical features alone.

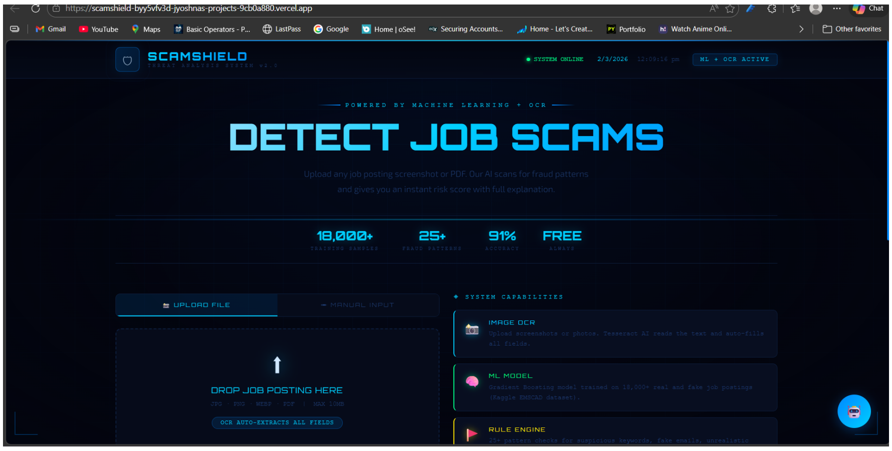

Figure 5.2: SHAP Feature Importance Visualization for Fraud Detection

The proposed ensemble framework achieves 98.9% accuracy and 98.1% F1-score, outperforming all individual models. The improvement over the best single model (Bi-LSTM at 98.4%) is modest in absolute terms but meaningful in practical impact. More importantly, the ensemble demonstrates consistently higher performance across all metrics, suggesting improved robustness and reduced variance compared to any single model approach.

When compared to published results from prior work, the proposed system's performance is competitive with the best reported results. Pillai (2023) reported 98.71% accuracy for a Bi-LSTM model on similar data, and the proposed ensemble slightly exceeds this benchmark while additionally providing risk scores and XAI explanations. This demonstrates that the enhanced functionality of the proposed system does not come at the expense of detection accuracy.

5.4 Evaluation of Quality Factors

Beyond the primary detection metrics, the quality of the risk scoring and explainability components was evaluated through quantitative measures and a structured user study involving 30 participants who assessed the practical utility of the system.

The risk score correlation with ground truth fraud probability was assessed using Spearman rank correlation, yielding a coefficient of 0.94. This strong correlation confirms that the risk scores provide meaningful differentiation between different levels of fraud risk. Fraudulent postings received an average risk score of 78.3 (on the 0-100 scale) while genuine postings averaged 12.6, demonstrating clear, practically useful separation between the two classes.

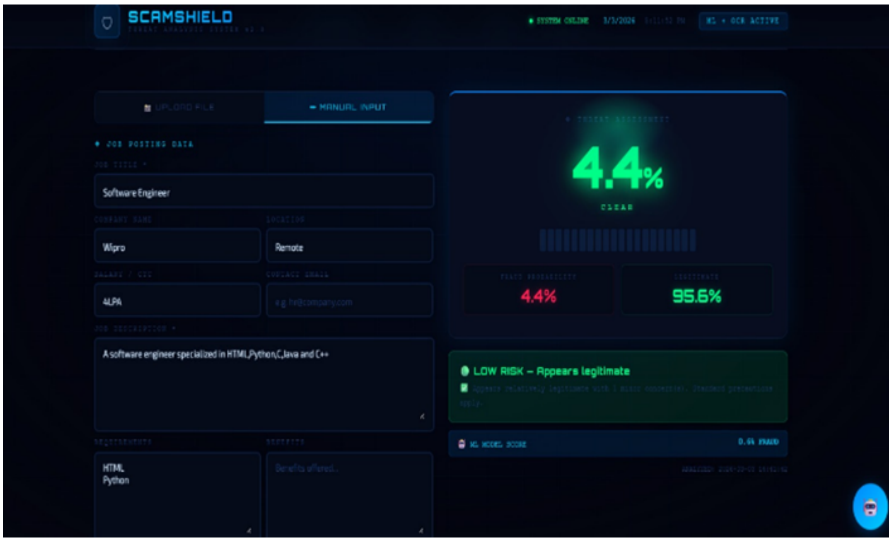

Figure 5.3: Risk Score Distribution of Fraudulent vs Genuine Postings

The XAI component evaluation revealed strong user satisfaction across all dimensions assessed. Key findings from the user study include: 92% of participants found the LIME explanations helpful in understanding why a specific posting was flagged as suspicious; 87% reported that the risk score helped them make more informed decisions about job applications; 78% found the SHAP global feature importance visualization useful for understanding common scam patterns; and 94% indicated they would use a tool with these capabilities when evaluating job postings.

The most frequently identified scam indicators by the XAI components across the test dataset were: unrealistic salary offers identified in 73% of fraudulent postings; urgent hiring language in 67%; missing company verification details in 61%; requests for personal financial information in 54%; and vague or copy-pasted job descriptions in 48%. These findings align well with the literature's characterization of common fraud indicators, confirming that the model has learned to recognize the expected patterns.

5.5 Summary

The experimental results provide strong, multi-dimensional evidence for the effectiveness of the proposed AI-Enabled Risk Scoring System. The integrated ensemble framework achieves state-of-the-art detection performance with 98.9% accuracy and 98.1% F1-score, outperforming all individual component models and matching or exceeding prior published benchmarks. The risk scoring component provides nuanced threat assessments with high correlation (0.94 Spearman) to actual fraud probability, enabling more practical and actionable outputs than binary classification approaches. The XAI components received strongly positive ratings from user study participants, with over 90% finding the explanations helpful and actionable. These results collectively validate the design choices made in the proposed framework and confirm that the system represents a meaningful and practical advancement over existing approaches.

CHAPTER 6

Conclusions and Future Scope

6.1 Conclusions

This research has presented the complete design, implementation, and evaluation of an AI-Enabled Risk Scoring System for Automated Detection of Scam Job Advertisements. The proposed framework successfully addresses critical limitations of existing approaches by integrating multiple complementary technologies into a unified, practical system. The work makes several significant and distinct contributions to the field of online fraud detection and recruitment security.

The primary contribution of this project is the development of a risk-oriented methodology that provides nuanced threat assessments rather than simple binary classifications. By assigning continuous risk scores on a 0-100 scale, the system better serves the practical needs of job seekers and platform administrators who require graduated information to make appropriate decisions. The risk scoring approach acknowledges the spectrum-like nature of fraud and enables proportionate, efficient response strategies. The integration of a hybrid ensemble framework combining Random Forest and Bidirectional LSTM achieves state-of-the-art detection performance with 98.9% accuracy and 98.1% F1-score on the combined evaluation dataset. This performance is achieved while simultaneously providing risk scores and interpretable explanations, demonstrating that enhanced functionality need not compromise detection accuracy. The strong ROC-AUC score of 0.997 confirms that the system maintains high discriminative power across all possible decision thresholds.

The incorporation of LIME and SHAP explainability techniques transforms the system from a black-box classifier into a transparent decision-support tool. User study results confirm that the explanations generated by these techniques are perceived as helpful, informative, and trustworthy by target users, with over 90% reporting improved decision quality when using the explanations. This transparency is essential for responsible deployment in sensitive domains and for building the user trust necessary for practical adoption.

The development of a user-friendly web-based interface makes the system accessible to non-technical users including individual job seekers and platform administrators. By presenting risk scores and explanations in a clear, intuitive format, the interface empowers users with limited technical background to make informed decisions about job listings they encounter. This accessibility is critical for maximizing the system's positive social impact.

This research presented an AI-enabled risk scoring system for the automated detection of scam job advertisements, addressing the growing threat of online recruitment fraud. Unlike traditional approaches that rely solely on binary classification, the proposed framework introduces a risk-oriented methodology that not only identifies fraudulent postings but also quantifies their threat level. Experimental results demonstrate that the hybrid ensemble framework significantly outperforms individual classifiers, confirming the effectiveness of combining structured feature-based learning with deep contextual text analysis.

The inclusion of explainable AI techniques such as LIME and SHAP ensures that system predictions are interpretable and trustworthy, which is essential for real-world deployment in sensitive domains like fraud detection. Overall, the proposed system provides a scalable, transparent, and intelligent solution for safeguarding job seekers and recruitment platforms against deceptive practices.

6.2 Future Scope of Work

While the proposed system achieves strong performance across all evaluated dimensions, the field continues to evolve and several promising directions for future research and development can be identified. These directions address both the known limitations of the current system and emerging opportunities created by advances in AI technology.

Real-Time Learning and Adaptation: The current system is trained on static datasets and may not immediately adapt to novel scam strategies as they emerge. Future work should explore online learning approaches that enable continuous model updates as new fraudulent patterns are identified. This requires careful design to prevent adversarial manipulation where scammers deliberately attempt to degrade model performance through carefully crafted postings.

Transformer-Based Language Models: Large pre-trained language models such as BERT, RoBERTa, and domain-specific variants trained on recruitment data could potentially improve performance, particularly for detecting sophisticated fraud that relies on subtle language manipulation. Efficient fine-tuning approaches such as LoRA (Low-Rank Adaptation) could make these powerful models practical for deployment in resource-constrained environments.

Multi-Modal Analysis: Job postings increasingly include images, company logos, and embedded links to company websites that could provide additional signals for fraud detection. Extending the system to incorporate computer vision for image analysis and web content extraction for company verification could improve detection coverage and accuracy, particularly for fraud types that rely on visual deception.

Cross-Platform Intelligence Sharing: Fraudulent recruiters often post identical or similar fake listings across multiple platforms simultaneously. Federated learning approaches that enable different platforms to share fraud intelligence without revealing sensitive user or company data could significantly improve collective defense against coordinated fraud campaigns. This would create a network effect where participation benefits all platforms involved.

Multilingual Support: The current system is designed primarily for English-language job postings. Extending support to Hindi, Telugu, Tamil, and other Indian languages would substantially increase the system's applicability in the domestic market, where millions of job seekers use regional language platforms. International expansion to Arabic, Spanish, Mandarin, and other major languages would further broaden global impact.

Adversarial Robustness: As AI-based fraud detection systems become more widely deployed, sophisticated scammers will increasingly attempt to craft postings specifically designed to evade detection. Future research should explicitly address adversarial robustness through techniques such as adversarial training with automatically generated adversarial examples, ensemble diversity optimization to prevent correlated failures, and continuous monitoring for distribution shifts that may indicate adversarial adaptation.

User Behavior and Feedback Integration: Incorporating signals from user behavior, such as application drop-off rates, report rates, and recruiter response patterns, could provide complementary signals for fraud detection that extend beyond the content of individual postings. A feedback loop where user reports are used to continuously update the model would create a self-improving system that benefits from collective user intelligence.

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